

FRANCESCA ROSE T. BEJENIA

Computer Engineer

☎ 09621692605 | ✉ frbejenia@gmail.com | 📍 Brgy. Limao, Samal City

Professional Summary

A motivated Bachelor of Science in Computer Engineering graduate from Holy Cross of Davao College, with hands-on experience in hardware and software development through academic projects and training, and a strong foundation in problem-solving and system design. Proven leadership and organizational experience through active involvement in student leadership and technical organizations. As a fresh graduate, I am eager to learn, apply and further develop technical knowledge, contribute to practical solutions, and grow professionally within a dynamic organization.

EDUCATION

Bachelor of Science in Computer Engineering
Holy Cross of Davao College | A.Y. 2022–2026
Citation Awardee

Senior High School — General Academic Strand
Limao Senior High School | S.Y. 2020–2022
With High Honors

TESDA: ENTERPRISE-BASED EDUCATION AND TRAINING

Web Development with Bubble.io
Completed

WORK EXPERIENCE / OJT

Technical Writer / Embedded Systems Intern
CreateLabz (Cloudline Technologies Inc.)

- Developed and documented a Smart Plant Monitor with automated watering, integrating soil moisture sensing, OLED and 8×8 LED matrix displays, motor/relay control, and PWM-based pump regulation.
- Created technical tutorials for an Intelligent Transportation Kit, documenting five projects involving sensors, servos, RFID, LCDs, and LED displays.
- Developed and documented a Smart Home Kit featuring seven security and home-automation projects, including earthquake and fire alarms, automatic door control, password access, drying rack, pet feeder, and automated watering systems.
- Produced public-facing technical documentation covering hardware setup, wiring diagrams, source code, and code explanations for publication on the company's tutorials page.

RESEARCH DESIGN PROJECT

SeedSpark — Autonomous Electro-Photosynthetic Microclimate Chamber

- Contributed to the design and development of an autonomous embedded system for tomato seedling cultivation, integrating a microcontroller with environmental sensors for temperature, humidity, soil moisture, soil pH, NPK, and CO_2 monitoring.
- Worked on the integration of automated subsystems including irrigation, cooling, LED grow lighting, ESP32-CAM monitoring, and bioelectrical root stimulation for controlled seedling growth.
- Developed the companion mobile application with Firebase integration for real-time environmental monitoring, remote system control, notifications, and historical data visualization.
- Contributed to firmware development, hardware integration, troubleshooting, testing, and calibration, helping refine the system from initial design through a functional working prototype.

TECHNICAL SKILLS

- **Programming:** Basic knowledge of C/C++, Java, Python, and TypeScript
- **Embedded Systems:** Arduino, ESP32/ESP8266, Sensor & Actuator Integration
- **Hardware:** PCB Etching & Layout Design, Schematic & Circuit Design, Circuit Simulation, Breadboarding
- **Software Development:** Dart/Flutter, Firebase, Git
- **Web Development:** Buble.io (No-Code Web App Development)
- **Networking:** Basic Networking
- **Tools:** VS Code, Arduino IDE, Microsoft Office, Google Workspace, Tinkercad, Proteus, Eagle, Circuit Wizard

ACHIEVEMENTS & TECHNICAL ACTIVITIES

- 1st Runner-Up, Best Paper Presentation, CET Conversazione (2026)
- Outstanding Officer, Institute of Computer Engineers of the Philippines Student Edition – Region XI (A.Y. 2023–2024)
- I Inspire Awardee, Holy Cross of Davao College (A.Y. 2024–2025)
- Champion, Quiz Bowl, CET Technofair (2024)
- Dean's Lister, Holy Cross of Davao College (A.Y. 2022–2023)
- Facilitator, Arduino for Beginners Seminar-Workshop for 2nd Year ECE & CpE Students (March 2025)
- Facilitator, PCB Fabrication Seminar on PCB Layout & Hardware Design for Freshmen ECE & CpE Students (October 2024)

LEADERSHIP & AFFILIATIONS

- External Vice President, Alliance of Computer Engineering Students (2024–2025)
- Assistant Treasurer, ICpEP.SE Region XI Chapter & Alliance of Computer Engineering Students (2023–2024)
- President, Supreme Student Government (2021–2022)

STRENGTHS

- Problem-Solving
- Analytical Thinking
- Technical Writing & Documentation
- Technical Troubleshooting
- Adaptability & Quick Learning
- Teamwork & Collaboration